MEHDI TAZI



Data & AI Architect | Chief Technical Officer

Data & AI Strategy & Consulting | On-premise & Cloud Platforms
Solution & Enterprise Architecture | AI Engineering

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Demystify the data realm

Book Author

- Co-author, The Definitive Guide to Data Integration — Packt, 2024.

View the book on Amazon

Education and Certifications

2019

- AWS Big Data – Specialty and Solutions Architect – Associate certifications.

2016

- PhD research at Paris-Est in Internet of Things (IoT) virtualization (in progress).
- Hortonworks Hadoop Administrator certification.
- Fundamentals of Data Science.

2014–2015

- Java OCAJ and Spring Core certifications.
- Apache Cassandra for Java Developers certification.

2013

- Certified ScrumMaster — Scrum Alliance.
- MongoDB for DBAs and MongoDB for Java Developers certifications.

2011

- Master's degree in MIAGE (Business Informatics), Engineering of Distributed Information Systems — ESIAG, Paris, France; with honors.

2010

- Engineering degree in Software Engineering — Fez, Morocco; with honors.
- Cisco CCNA certification — Rabat, Morocco; with honors.

Languages

English — Fluent.

French — Bilingual.

Arabic — Native language.

Technical and Architectural Expertise

Solution Design and Modeling

- Gather business requirements and define functional and technical solutions. Produce HLD/LLD for applications, infrastructure and integration flows.
- Model solutions with UML and Merise; document class, sequence, component, deployment, infrastructure and flow views to explain architecture choices.

Data and AI Platform Architecture

- Advise on data platforms, Lakehouses, Data Mesh, data lakes and labs; define blueprints aligning business priorities, ingestion, processing and serving.
- Design ingestion, processing and serving layers with Bronze/Silver/Gold medallion patterns. Define staging, exchange, logging and archive zones, and workspace boundaries.
- Plan platform and catalog migrations through phased coexistence and transitions, for example AWS to Databricks and AWS Glue Data Catalog to Unity Catalog.

Cloud Architecture and Data Engineering

- Architect on-premises, hybrid and multicloud solutions across AWS, Azure and Databricks, covering application and infrastructure requirements.
- Advise on, architect and develop scalable, distributed and parallel Big Data/NoSQL solutions.
- Deliver data engineering in Java, Scala and Python using Airflow, Spark, dbt, AWS (EMR, Glue, Athena and Lambda) and Databricks.

AI Application Design

- Design AI platforms and ML, LLM and agentic applications; define model, tool and data interactions and guide teams through use-case implementation.
- Integrate systems, BI, analytics and data science with data platforms; define governed access, scalable compute and monitoring for AI workloads.

AI Engineering and Industrialization

- Translate ideas and needs into software specifications, implementation plans and testable acceptance criteria using AI-assisted development workflows.
- Build harnesses with reusable context, instructions and automated validation; use AI-assisted coding, testing and pull-request automation, with human review.
- Automate releases, operation and maintenance through DevOps, infrastructure as code, MLOps and engineering stacks; improve software through monitoring and feedback.

Governance, Security and Reliability

- Establish technical and business catalogs, lineage and shared data dictionaries; define data quality plans, controls and access provisioning.
- Define sensitive-data and PII strategies and authorization models; implement observability tools, monitoring and security, architecture and technical audits.
- Establish business recovery and continuity plans (BRP/BCP). Design highly available, responsive, real-time, secure and governed solutions.

Enterprise Architecture and Leadership

Enterprise Architecture and IT Strategy

- Assess business needs and technology trends to develop and maintain IT strategy and an enterprise architecture vision; select, design and support solutions for business transformation.
- Define Data & AI strategy and roadmaps aligned with business needs and executive priorities.
- Align application, infrastructure and data architecture with target platform blueprints and migration roadmaps; advise Data, Architecture and IT leadership on choices and implementation priorities.

CTO, Consulting and Tender Responses

- Lead technology direction and develop the Data & AI strategy, consulting portfolio and service offering in line with client needs.
- Develop new business by identifying client needs, qualifying opportunities and shaping architecture and Data & AI consulting engagements.
- Lead responses to calls for tenders and RFPs, including technical proposals, solution design, delivery approaches, workload estimates and technical specifications.
- Recruit and develop technical teams; manage internal fixed-price projects from scope and estimates through staffing, planning, delivery monitoring and acceptance.
- Guide architecture decisions, oversee solution quality and support delivery teams across client engagements.
- Deliver technical keynotes and training.

Target Operating Model and Advisory

- Define the platform organization, team structure, roles and responsibilities, responsibility matrices and delivery backlogs for each team.
- Connect the target architecture to the organization that will operate it, and guide teams on delivery and operating priorities.

Project Management Skills

- Communicate goals and vision, and prepare reports for executive management.
- Assess resource availability and maintain delivery capacity plans.
- Coach and lead teams; estimate, plan, organize and monitor projects.
- Break products down into use cases for release planning, then divide use cases into tasks for iteration planning.
- Prepare releases and iterations; maintain user stories and the backlog with the product owner.
- Identify and manage risks.
- Facilitate meetings, motivate teams and strengthen collaboration within and across teams.
- Support teams in setting up and onboarding their use cases.
- Produce delivery indicators, including burnup/burndown charts, team velocity, task tracking, code quality and scope monitoring.

FinOps Skills

- Design and implement FinOps governance, standards and best practices.
- Prepare cost forecasts and financial estimates.
- Participate in architecture committees to guide projects from the design phase toward cost-efficient architecture choices.
- Optimize cloud service purchasing.
- Optimize infrastructure by right-sizing machines and clusters.
- Optimize technical costs through engineering patterns such as bucketing, partitioning and Z-ordering; recommend lightweight technology stacks and decision trees for technology selection.
- Optimize storage through storage tiering and data lifecycle management.
- Implement FinOps solutions to identify and eliminate orphaned resources, using automatic tagging and a configuration management database (CMDB).
- Automate FinOps activities with in-house tools, including Tableau and Power BI dashboards and a CMDB.
- Monitor usage and spend, and provide regular reporting with KPIs.
- Manage FinOps communities and promote FinOps practices, culture and tools through webinars, workshops and events.

Technology Stack

Data and AI platforms	Databricks · Native AWS / Azure data platforms · Cloudera Private Cloud Base
Ingestion and messaging	Azure Data Factory · AWS DMS · Amazon Kinesis · Azure Event Hubs · Kafka · Confluent / Aiven · Talend · Flume · Databricks Auto Loader · Lakeflow Connect · Amazon SNS · FTP
Processing and transformation	Apache Spark · Hadoop · AWS Glue ETL · Amazon EMR · AWS Glue DataBrew · dbt · Lakeflow Spark Declarative Pipelines · Databricks Runtime / Photon
Orchestration and job scheduling	Airflow · Lakeflow Jobs · AWS Batch
Storage and databases	Amazon S3 · S3 Glacier / Deep Archive · ADLS Gen2 · HDFS · PostgreSQL · Amazon RDS · DynamoDB · Cosmos DB · Cassandra · MongoDB · HBase · Redis
Table and file formats	Delta Lake · Parquet · Avro · CSV · XML · JSON · YAML
SQL and data serving	Databricks SQL / SQL Warehouses · Amazon Athena · Hive · Impala · Trino / Presto · Amazon API Gateway · REST / GraphQL APIs
Analytics and AI	Power BI · Tableau · Amazon QuickSight · Dataiku · Amazon SageMaker AI
Catalogs and sharing	Unity Catalog · AWS Glue Data Catalog · Collibra · OpenSharing (Delta Sharing)
Security and networking	LDAP / OpenLDAP · Kerberos · HAProxy · AWS CloudTrail
Observability and operations	AWS CloudWatch · Splunk · Ambari · ZooKeeper · ServiceNow
Compute and containers	Amazon EC2 · AWS Lambda · Amazon EKS / ECS · Docker · Kubernetes
Delivery and automation	Git · Azure DevOps · Jenkins · Ansible · AWS CloudFormation · Terraform (in progress)
Languages	Python · SQL · Scala · Java / JEE · C# · CQL · JavaScript · HTML5 / CSS3
Development and quality tools	Jupyter · IntelliJ IDEA · Visual Studio · Maven / sbt · Sonar · Jira · QC · JHipster · SILVA
Operating systems	Windows · Unix · macOS

Industry Knowledge

Industry sectors: retail and e-commerce, energy, financial markets, retail banking, real estate, travel and business travel.

Professional Experience

Decathlon



Principal Data Architect

5 YEARS

Core Data Architect and Data Lake Architect.

- Supported the AWS-to-Databricks migration and defined a phased catalog strategy: AWS Glue Catalog/**Unity Catalog** coexistence, followed by full Unity Catalog adoption.
- Designed the target Lakehouse/Data Mesh architecture, HLD/LLD and strategic roadmap, covering governance, observability, PII and the **target operating model**.
- Contributed to the evolution of the platform towards Data & AI capabilities, defining architecture requirements for governed data access, scalable compute and operational monitoring.
- Supported core platform implementation for data management and analytics; qualified requirements, proposed solutions and guided teams through use-case delivery.
- Implemented automated platform provisioning and access management, and capabilities to manage and protect PII and sensitive data.
- Contributed to **FinOps** optimization through architecture reviews and technical efficiency improvements.
- Contributed to extending the platform across multiple cloud environments.
- Integrated Databricks with Collibra, AWS services and third-party technologies.

TECHNOLOGIES AWS · Databricks · Unity Catalog · dbt · Amazon EMR · AWS Glue · Amazon Athena · AWS Lambda · Amazon EKS / Kubernetes · Collibra

Professional Experience

ENGIE

Principal / Lead Big Data & Cloud Architect — Applications & Infrastructure

4 YEARS

ENGIE GEM and ENGIE IT | Paris, France | via IT INSIGHT

Architecture studies and delivery across data platforms, applications and cloud infrastructure.

- Owned **Big Data architecture** across on-premises, Azure and AWS hybrid and multicloud environments.
- Implemented automated controls for inefficient cloud implementations and partitioning errors; automated billing, monitoring and user/group administration, and optimized application cost and performance.
- Evolved the **data strategy** and enterprise architecture vision; assessed technologies and business needs, designed solutions and advised on architecture patterns for high-performance use cases.
- Managed development teams, onboarded use cases and worked with business/IT stakeholders on data opportunities, guidance and delivery approaches.
- Coordinated application, infrastructure, API, security and DevOps architects to maintain a consistent data offering.
- Worked with ENGIE Group on shared data products, SDKs, catalogs, data services, billing and the operating organization across entities.
- Created code samples, wikis, how-to guides, frameworks, workshops and best practices for onboarding, development and Big Data/cloud skills.
- Implemented APIs over the data lake, governance, security, optimization and automation; integrated Collibra, Dataiku, Power BI, authentication and Splunk, with DevOps and infrastructure as code.
- Industrialized log analysis for predictive maintenance and monitoring, and proposed architectures aligned with business requirements.

TECHNOLOGIES Databricks · AWS · Apache Spark · AWS Glue · Amazon S3 · Amazon Athena · Amazon Kinesis · Amazon SNS · AWS DMS · Amazon SageMaker · Amazon DynamoDB · AWS Lambda · Docker · Amazon EC2 · Amazon API Gateway · AWS CloudFormation · Amazon CloudWatch · AWS CloudTrail · AWS Batch · Amazon EKS · Azure · Azure Data Factory · Azure Data Lake Storage · U-SQL · Azure HDInsight · Azure Event Hubs · Scala · Python · Hadoop · Collibra · Dataiku · Power BI · Splunk

AXA

Big Data & NoSQL Architect — Applications & Platform

2 YEARS

AXA World Tech and Data Innovation Lab | via IT INSIGHT

Delivered Big Data architecture projects and studies, and maintained AXA's Big Data cluster.

- Produced HLD/LLD and **technical architecture documentation**, including class, sequence, component, deployment, infrastructure and flow diagrams; initiated and led projects and proofs of concept.
- Evaluated Big Data solutions and Kafka–Navigator integration; led Cloudera Navigator for governance and access monitoring.
- Designed the SelfService architecture to automate platform operations from ticketing through Ansible scripts; developed deployment and operational automation with Ansible and Python.
- Administered clusters and provided third-level support for CDH5, HDFS, Hive, Spark and Oozie.
- Integrated platform tools with relational databases, logs, FTP, LDAP and Kerberos, using CDH5, OpenLDAP, CentOS and PostgreSQL.

TECHNOLOGIES Cloudera Hadoop · Ansible · LDAP · Kerberos · HDFS · Hive · Kafka · Spark · Oozie · Flume · Python · Java · SILVA · HAProxy · Unix · PostgreSQL · Docker · Cloudera Navigator · CDH5 · OpenLDAP · CentOS

Professional Experience

BI-NEWVISION

Chief Technical Officer (CTO) & Principal Data Architect

7 YEARS

Technology leadership and Data & AI consulting

Lead BI-NEWVISION's technology strategy and Data & AI consulting offering, combining company leadership with direct architecture advisory.

- Define and evolve the Data & AI strategy, consulting portfolio and service offering, aligning technical capabilities with client needs.
- Develop new business by qualifying opportunities and shaping architecture and Data & AI consulting engagements.
- Lead tenders and RFP responses, including solution design, technical proposals, delivery approaches and workload estimates.
- Set technical direction across engagements, guide architecture decisions and oversee solution quality.
- Recruit, lead and support delivery teams; manage internal fixed-price projects, coordinate priorities and provide architecture advisory on client missions.

OCTO Technology

Big Data & NoSQL Architect / Agile Coach — Applications

2 YEARS

BNP PARIBAS, LA POSTE, COLISSIMO, EDF-DOAAT, ERDF, AL BARID BANK, SMCP, OCTO ACADEMY

Delivered multiple consulting projects, primarily as a Big Data Architect and occasionally as an Agile Coach.

- Scoped business needs and Big Data use cases, established roadmaps and MVPs/proofs of concept, and managed implementation projects.
- Provided hands-on consulting across administration, architecture, governance and security; designed technical, conceptual, functional, organizational and methodological architectures.
- Designed **data lakes and data labs**, covering organization, modeling and governance; integrated existing systems, BI and analytics with Big Data platforms.
- Designed ingestion, exchange, logging, archiving, modeling and serving layers.
- Established BRP/BCP and designed highly available, responsive, real-time and secure solutions.
- Designed, managed and developed Big Data solutions; automated scripts, infrastructure, deployments and configuration.
- Supported presales, delivered Big Data training and provided Agile/Scrum coaching.

TECHNOLOGIES Java / JEE · Scala · Hadoop · Spark · Kafka · Ansible · Docker · Cassandra · HBase · Hive · Sqoop · Parquet · Avro · HDFS · Ambari · ZooKeeper · Confluent.io · Jupyter · Git · Jenkins · JHipster

References and Publications

References

Thierry GRIMA

Group Chief Analytics Officer

ENGIE & AXA

linkedin.com/in/thierrygrima/

Raphael MANSUY

Data Architect

DECATHLON

linkedin.com/in/raphaelmansuy/

Publications and Public Speaking

- Co-author, The Definitive Guide to Data Integration — Packt, 2024.

View the book on Amazon

- Reviewer of MongoDB for Java Developers — Packt Publishing.

goo.gl/YxskzV

- Co-organizer of the Casablanca MongoDB User Group.
- Internet of Things Virtualization poster for the NIST Cloud Computing Forum & Workshop IX — National Institute of Standards and Technology, USA.
- PhD research paper on IoT virtualization: “A Micro-Clouds Sensors Virtualization and Composition Approach for the Internet of Things.”
- Facilitator and speaker for the “Games Programming” workshop at JNJD, INPT, Morocco.
- Keynote speaker on technical topics and Scrum management.
- Creator and contributor to open-source projects: Chimera2D, MoovinParis and Itefacio.com.
- Author of technical articles on Medium.

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